

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	5 MARKS

CHAPTER – IV PROJECT DESIGN

4.1 – Problem Solution Fit

The increasing demand for electricity and the lack of detailed insights into consumption patterns make it difficult for households, industries, and policymakers to manage energy resources efficiently. Variations in electricity usage across regions and seasons, along with the inability to identify peak demand periods, often lead to energy wastage, higher operational costs, and challenges in planning future power requirements. To address these issues, the project "Plugging into the Future: An Exploration of Electricity Consumption Patterns" analyzes electricity consumption data collected from households, industries, commercial sectors, and smart meters. By applying data preprocessing, trend analysis, and visualization techniques, the system identifies usage patterns, seasonal variations, and peak demand periods. Historical data is also utilized to gain deeper insights into consumption behavior.

The proposed solution provides interactive dashboards, region-wise and time-wise analysis, and comprehensive reports that support informed decision-making. These insights help energy providers optimize resource allocation, enable consumers to adopt energy-efficient practices, and assist policymakers in forecasting future electricity demand. Ultimately, the project contributes to sustainable energy management and promotes efficient utilization of electricity resources.